	Introduction to Robotics		
Course Code	IE410	Credit Structure	3-0-2-4 (L-T-P-C)
Category	Core / Elective	Semester	Winter Semester (AY 24-25)
Program	Introduction to Robotics is an ICT and Technical Elective course for students of BTech ICT, RAS (**R**obotics and **A**utomation **S**ystems) minor and Honours programs. It is also offered as an elective for MTech program.		
Prerequisites	-		
Course Objectives/ Brief Course Description	This course provides a comprehensive introduction to the exciting and interdisciplinary field of robotics, combining theory and hands-on practice to enable students to build and program robotic systems from scratch. Students will gain foundational knowledge in key topics such as coordinate transformations, kinematics, dynamics, control, and robotics graphics. The course covers advanced concepts like trajectory planning, sensor integration, and hybrid control strategies, preparing students for real-world applications. Through interactive lectures and laboratory sessions, students will design, implement, and analyze robotic systems, applying modern tools and programming techniques using software packages like MATLAB-Simulink and Python. By the end of the course, students will acquire practical skills and theoretical insights required for careers in robotics, automation, and related industries, while also addressing societal and ethical considerations in engineering solutions. This course is ideal for students seeking to explore robotics as a technical elective, with applications in academia and industry. **Course Objectives** 1. To provide students with foundational knowledge of robotics, including coordinate transformations, kinematics, dynamics, and control. 2. To enable students to design, build, and program robotic systems from scratch, emphasizing practical applications and theoretical principles. 3. To introduce students to sensors, actuators, and power management circuits essential for robotic systems. 4. To foster skills in robotic computing and simulation using tools like MATLAB-Simulink and Python for real-world robotic applications. 5. To prepare students for interdisciplinary challenges in robotics and automation, aligning with industry needs and societal development.		
Evaluation/ Grading Policy	Evaluation scheme: (*i*) First and Second In-Sem Exam: 15%+15%=30%, (*ii*) End-Sem Exam: 25%, (*iii*) Quizzes, Assignments, (Lab/Group) Projects, 15%+05%+15%=35%,		

	(*iv*) Labs: 10%, and (*v*) Attendance 0% (Zero %).
Course Materials/ References	1. **Introduction to Robotics: Analysis, Control, Applications**, 3rd Ed. Saeed B. Niku, Wiley, 2019. 2. **Robotics - Modelling, Planning and Control**, 1st Ed. B. Siciliano, L. Sciavicco, L. Villani, and G. Oriolo, Springer, 2009. 3. **Introduction to Robotics: Mechanics and Control**, 3rd Ed. John J. Craig, Addison-Wesley Publishing Company, 2003. 4. **Robotics, Vision and Control, Fundamental Algorithms in Python**, 3rd edition, Peter Corke, Springer, 2023. 5. **Introduction to Autonomous Mobile Robots**, 2nd Edition, Roland Siegwart, Illah R. Nourbakhsh, and Davide Scaramuzza, The MIT Press, 2011. 6. **Robot Manipulator Control: Theory and Practice**, Frank L. Lewis, Darren M. Dawson, and Chaouki T. Abdallah, CRC Press, 2003. 7. **Modern Robotics**, K.M. Lynch, Cambridge University Press, 2017. 8. Other resources like, notes, papers shared on Google Drive from time to time.
Detailed Course Content	APPENDIX

Course Outcomes

Upon successful completion of this course, students will be able to:

1. Demonstrate a clear understanding of the principles of robotic architecture, kinematics, dynamics, and robot control.

2. Apply knowledge of coordinate transformations and Denavit-Hartenberg (D-H) parameters to solve robotic problems involving forward and inverse kinematics.

3. Design and implement control strategies, including PID and hybrid control, for robotic applications.

4. Develop and integrate sensors, actuators, and power management systems to create functional robotic systems.

5. Exhibit proficiency in programming robotic controllers and simulations using MATLAB-Simulink, Python and other tools/packages.

6. Collaborate effectively in teams to design, build, and test robotic systems, fostering teamwork and leadership skills.

7. Analyze and solve interdisciplinary problems in robotics, demonstrating an understanding of societal and ethical implications.

8. Use modern tools and techniques to design and simulate robotic solutions for industrial and societal applications.

PO-COs Matrix/ Mapping Table:

PO/PSO	CO1	CO2	CO3	CO4	CO5	CO6	CO7	CO8
PO1 Engineering knowledge	✓	✓	✓	✓	✓		✓	✓
PO2 Problem analysis	✓	✓	✓	✓	✓		✓	✓
PO3 Design/development of solutions	✓	✓	✓	✓	✓		✓	✓
PO4 Conduct investigations of complex problems		✓	✓	✓				✓
PO5 Modern tool usage			✓	✓	✓			✓
PO6 The engineer and society							✓	
PO7 Environment and sustainability				✓			✓	
PO8 Ethics							✓	
PO9 Individual and team work						✓		
PO10 Communication					✓	✓		
PO11 Project management and finance				✓		✓		
PO12 Life-long learning	✓				✓		✓	✓
PSO1 Theoretical concepts and practical analysis	✓	✓	✓	✓	✓		✓	✓
PSO2 Hardware and software integration		✓	✓	✓	✓	✓		✓
PSO3 Social responsibility and real-world problem-solving			✓	✓	✓	✓	✓	✓

APPENDIX: Detailed Course Content (Session-wise/ Module-wise)

Planned Lecture Schedule

Sl. No	Description		No. of Lectures
1	**Basic of Robotics**		**4**
	1.1	History of Robotics	1
	1.2	Robot Anatomy	1
	1.3	Robot Links and Joints	1
	1.4	Robotics Graphics	1
2	**Robot Coordinate Systems**		**4**
	2.1	Cartesian and Polar Coordinate Systems	1
	2.2	Rotational and Translational Matrices (1-D, 2-D, and 3-D)	3
3	**Forward Kinematics**		**7**
	3.1	Passive and Active Transformations	1
	3.2	Homogeneous Transformation	1
	3.3	Euler Angles	1
	3.4	Quaternion	1
	3.5	Jacobian	1
	3.6	Denavit-Hartenberg (D-H) Parameters	2
4	**Inverse Kinematics**		**6**
	4.1	Inverse Kinematics (IK) in 2-D and 3-D	1
	4.2	IK: Cosine Method	1
	4.3	IK: Numerical Method	1
	4.4	Inverse Jacobian and Examples	2
	4.5	Redundancy and Singularities	1
5	**Sensor and Actuators**		**7**
	5.1	Various sensors useful in robotics (Pressure sensor, Gyro sensor, Accelerometer, GPS, Vision Sensor)	3
	5.2	Various Actuators, (DC Motor, Servo-Motor)	2
	5.3	Power Management and associated circuits	2
6	**Robot Dynamic and Force Analysis**		**4**
	6.1	Equations of Motion, Newton-Euler Formulation	2
	6.2	Lagrange Formulation and Robot Applications	2
7	**Trajectory Planning**		**4**
	7.1	Path vs. Trajectory	2
	7.2	Polynomial, Cartesian, and Smoothing Trajectories	2
8	**Robot Control**		**6**
	8.1	PD Control, PID Control	2
	8.2	Position Control, Force/Torque Control	2
	8.3	Hybrid and Advanced Control	2